SECP3723 SYSTEM DEVELOPMENT TECHNOLOGY

SEMESTER 1 2024/2025

SECTION 1

Individual Test: Course Registration System

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System Introduction

The course registration system is the individual project assigned for this subject, System Development & Technology (SECP3723). In this individual project, we have applied a few programming languages which are specialised in web application development such as PHP, CSS, JavaScript and HTML. By applying this knowledge learnt along the semester, a web-based course registration system was developed which provides students a platform to register for courses for upcoming semesters.

The course registration system has 3 types of users, which are admin, student and lecturer. The student is the user that can register for courses, view current and previous semester courses, search courses, modify their registration, cancel their registration, and edit their profile. The lecturer on the other hand can view their assigned courses for each semester, view student lists for each course they are assigned to, view course details and view student details. Finally, the admin is responsible to add new course details with the number of maximum students per course, modify existing course details, delete existing courses and amend or cancel registration.

System Design

Entity Relationship Diagram(ERD)

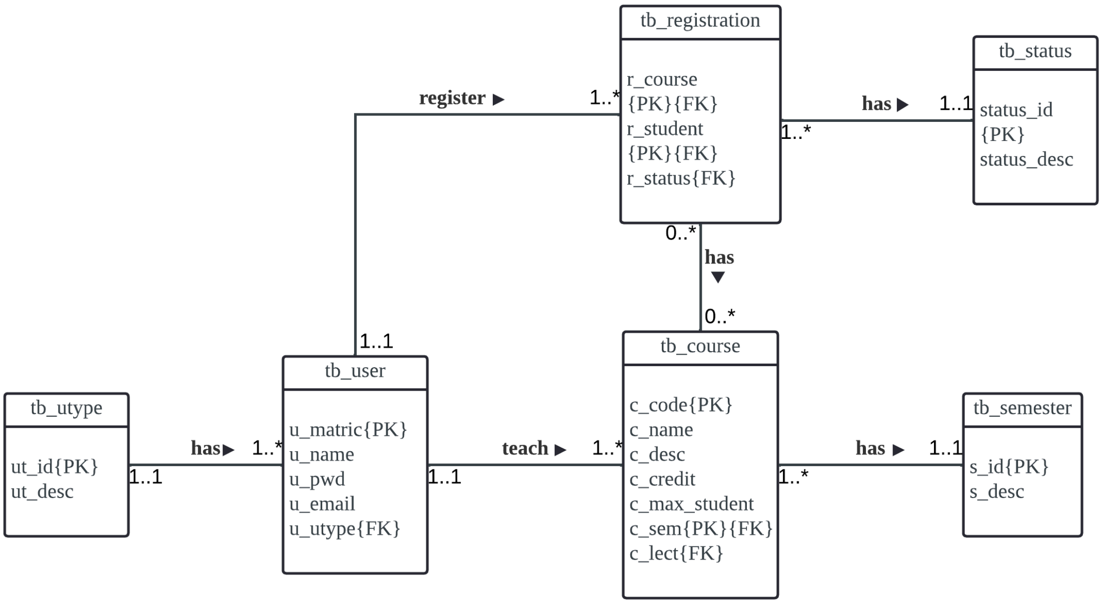


Figure 1.1 ERD of Course Registration System

Use Case Diagram

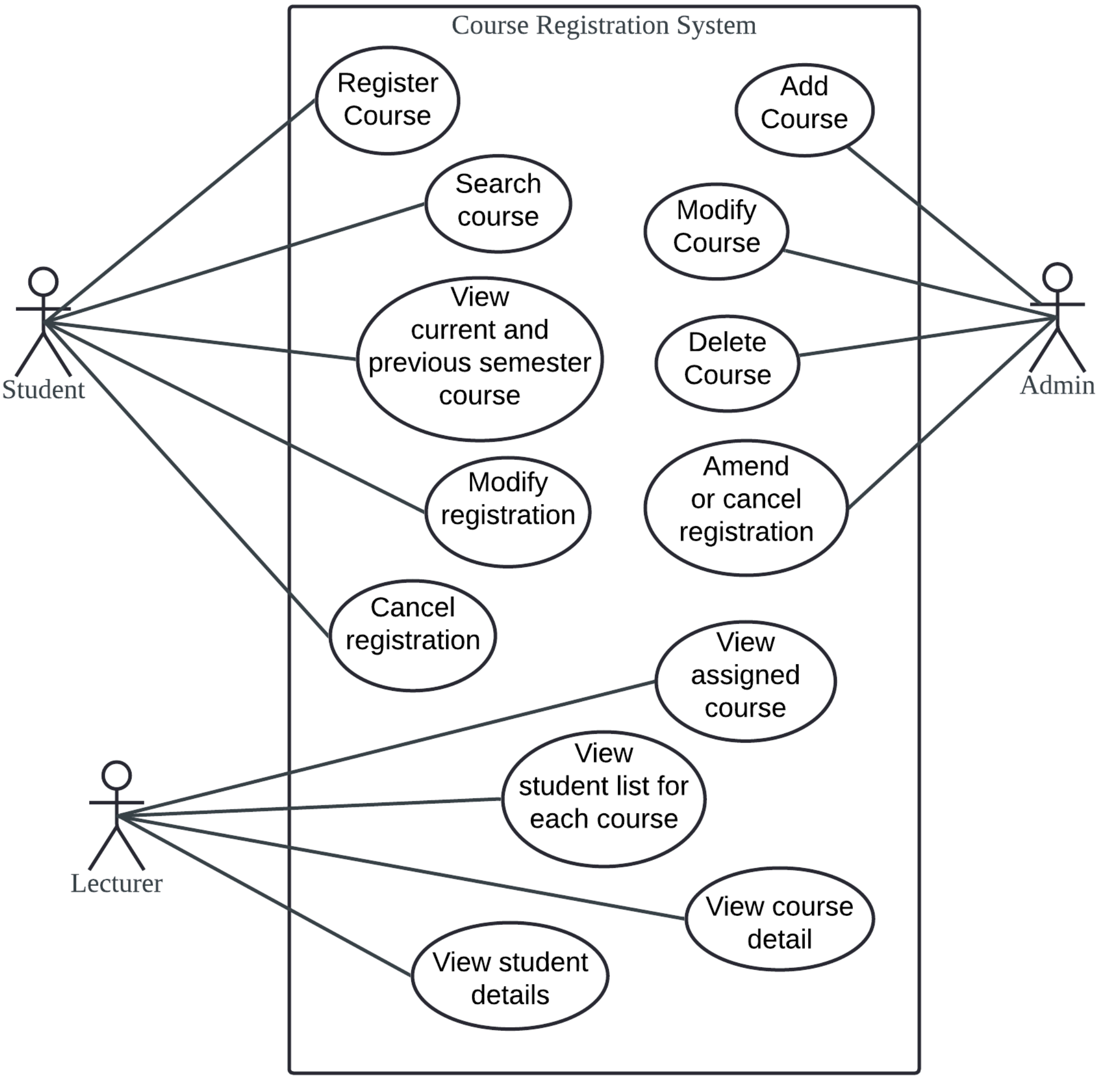


Figure 1.2 Use Case Diagram of Course Registration System

Development Software, Language, Technology and Tools

Software

The softwares that was used in this individual project to complete the car booking system are Visual Studio Code, XAMPP and Google browser. Visual Studio Code is a source code editor which supports a variety of programming and markup languages through the extensions that were provided for installation. Other than that, XAMPP is an open-source cross-platform web server solution stack package which is able to support me to host the course registration locally and offline by using Apache HTTP server. Lastly, the Google Chrome browser was used to test the functionalities of the system.

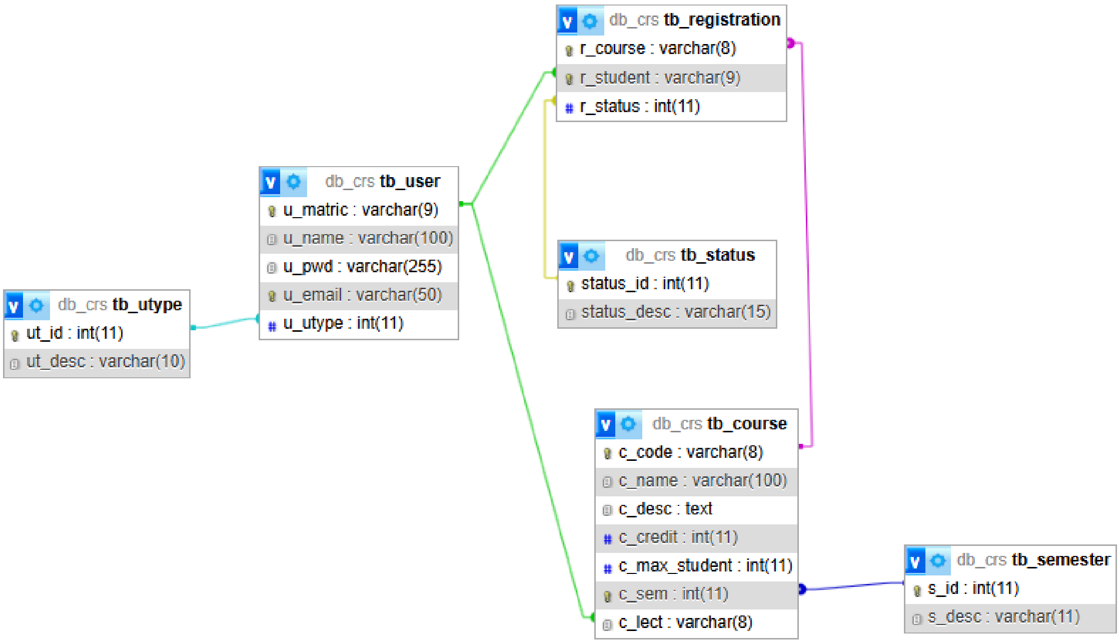
Languages

There are many types of programming languages that was was used to the develop the course registration system which were Hypertext Markup Language (HTML), Cascading Style Sheets (CSS), JavaScript, Hypertext Preprocessor (PHP) and Structured Query Language (SQL). HTML is a programming language that is used to build and structure web pages with the help of CSS to define the styles of the web pages. Other than that, JavaScript is a scripting language for web pages which use to trigger functions while certain events occur. PHP is a general proposed scripting language that connects the HTML to the database which is phpMyAdmin. Lastly, SQL language is also used in the phpMyAdmin to administrate the MySQL database of the system.

Tools

The tool used was Lucidchart which is an online tool that is used to construct the ERD and use case diagram of the system.

Database Design



Server: 127.0.0.1 » Database: db_crs » Table: tb_course

Browse | Structure | SQL | Search | Insert | Export | Import | Privileges | Operatic

Table structure | Relation view

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	**c_code**	varchar(8)	utf8mb4_general_ci		No	*None*			Change Drop More
☐	2	**c_name**	varchar(100)	utf8mb4_general_ci		No	*None*			Change Drop More
☐	3	**c_desc**	text	utf8mb4_general_ci		No	*None*			Change Drop More
☐	4	**c_credit**	int(11)			No	*None*			Change Drop More
☐	5	**c_max_student**	int(11)			No	*None*			Change Drop More
☐	6	**c_sem**	int(11)			No	*None*			Change Drop More
☐	7	**c_lect**	varchar(8)	utf8mb4_general_ci		No	*None*			Change Drop More

Server: 127.0.0.1 » Database: db_crs » Table: tb_registration

Browse | Structure | SQL | Search | Insert | Export | Import | Privileges | Oper

Table structure | Relation view

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	**r_course**	varchar(8)	utf8mb4_general_ci		No	*None*			Change Drop More
☐	2	**r_student**	varchar(9)	utf8mb4_general_ci		No	*None*			Change Drop More
☐	3	**r_status**	int(11)			No	*None*			Change Drop More

Server: 127.0.0.1 » Database: db_crs » Table: tb_semester

Browse | Structure | SQL | Search | Insert | Export | Import | Privileges | Operations

Table structure | Relation view

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	**s_id**	int(11)			No	*None*		AUTO_INCREMENT	Change Drop More
☐	2	**s_desc**	varchar(11)	utf8mb4_general_ci		No	*None*			Change Drop More

Server: 127.0.0.1 » Database: db_crs » Table: tb_status

Browse | Structure | SQL | Search | Insert | Export | Import | Privileges | Op

Table structure | Relation view

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	**status_id**	int(11)			No	*None*			Change Drop More
☐	2	**status_desc**	varchar(15)	utf8mb4_general_ci		No	*None*			Change Drop More

Server: 127.0.0.1 » Database: db_crs » Table: tb_user

Browse | Structure | SQL | Search | Insert | Export | Import | Privileges | Ope

Table structure | Relation view

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	**u_matric**	varchar(9)	utf8mb4_general_ci		No	*None*			Change Drop More
☐	2	**u_name**	varchar(100)	utf8mb4_general_ci		No	*None*			Change Drop More
☐	3	**u_pwd**	varchar(255)	utf8mb4_general_ci		No	*None*			Change Drop More
☐	4	**u_email**	varchar(50)	utf8mb4_general_ci		No	*None*			Change Drop More
☐	5	**u_utype**	int(11)			No	*None*			Change Drop More

Server: 127.0.0.1 » Database: db_crs » Table: tb_utype

Browse | Structure | SQL | Search | Insert | Export | Import | Privileges

Table structure | Relation view

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	**ut_id**	int(11)			No	*None*			Change Drop More
☐	2	**ut_desc**	varchar(10)	utf8mb4_general_ci		No	*None*			Change Drop More

Development Steps

Phase 1: Planning and Requirements Gathering.

In this phase, research was done to identify the requirements of the building. Questions like, “What type of data is involved in the system?”and “What sort of functionalities are needed to be in the system?”These sorts of questions help determine the system's purpose, objective, target audience, and range. The phase focuses on how things are done, identifying the proper flow of information throughout the system.

Phase 2: Designing the system

The system's ERD and use case diagram were first created. The ERD provides a visualization of the data needed in the system that also determines the logic and business rules of the system. The use case diagrams explain the context and requirements of the system showing ways in which a user interacts with the system. It also determines the scope of the system showing functionalities that are not covered by the system.

Phase 3: Developing the system

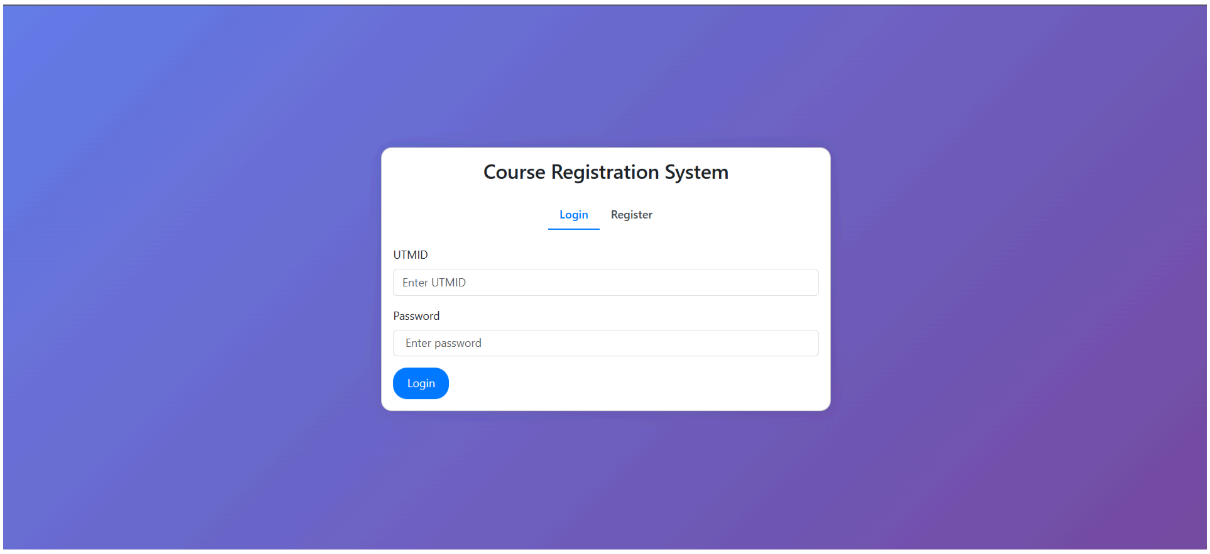
The system was developed with various programming languages that are used to create a functioning client side interface for presentation that is interactive while also having a server side that controls all the operations within the system such as CRUD operations, password encryption, SQL injection protection, password validation and resolving login error.

Phase 4: Testing the system

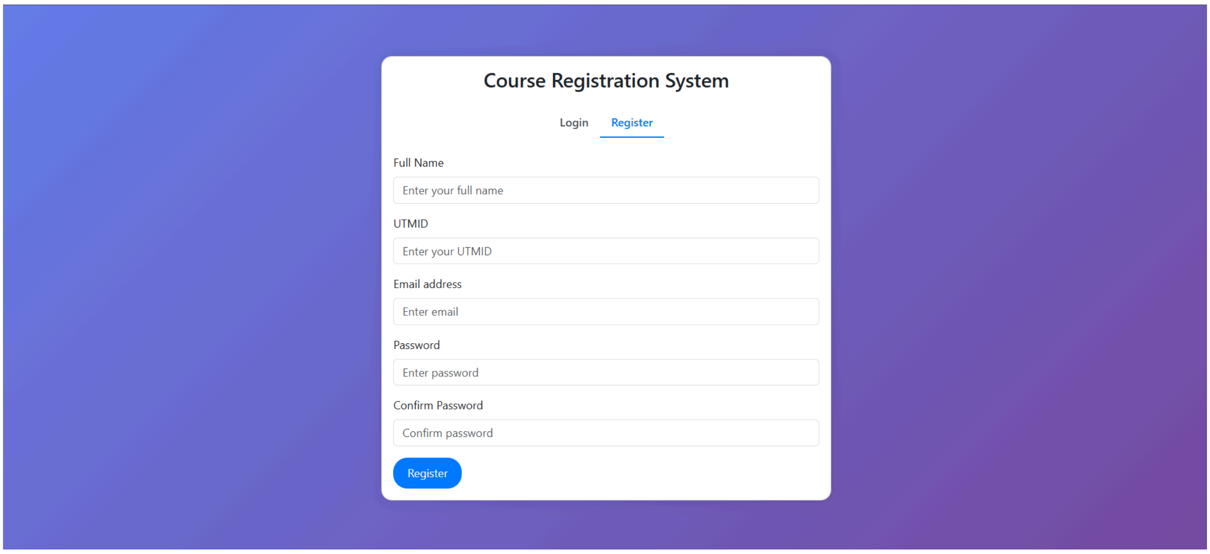
The testing phase was done each time a module or a use case was done. This phase is important to identify logical errors and bugs to ensure they can be fixed so that the system can work as intended. The testing was done on the web server each time the code was finished implementing. Then , an integration test was done by simulating the whole flow of the system to identify incompatibility issues for all the code that was developed.

System Interface

Login Interface for all user

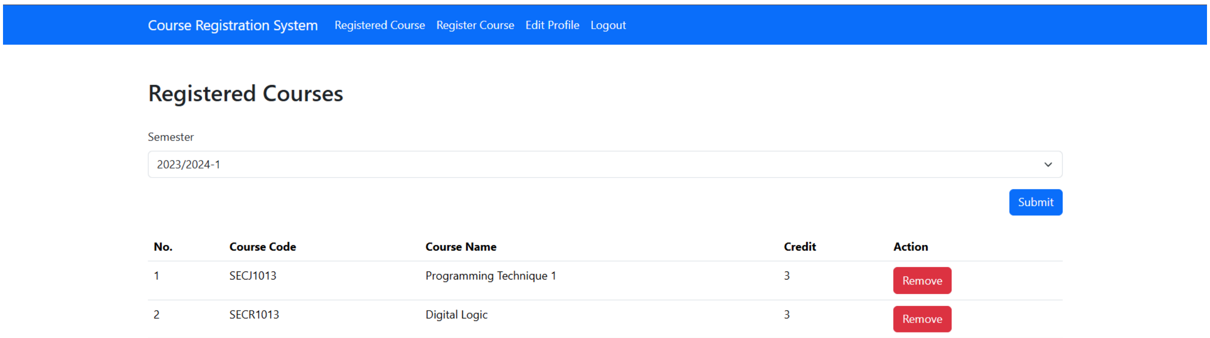


Registration Interface for Students



Student

Registered Course Page



Course Registration System Registered Course Register Course Edit Profile Logout

Registered Courses

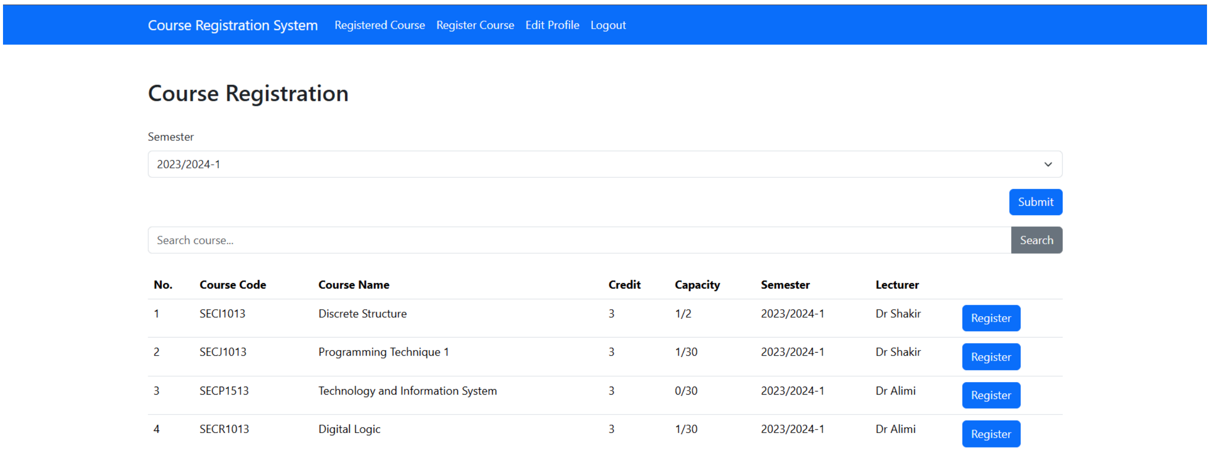
Semester

2023/2024-1

Submit

No.	Course Code	Course Name	Credit	Action
1	SECJ1013	Programming Technique 1	3	Remove
2	SECR1013	Digital Logic	3	Remove

Course Registration Page



Course Registration System Registered Course Register Course Edit Profile Logout

Course Registration

Semester

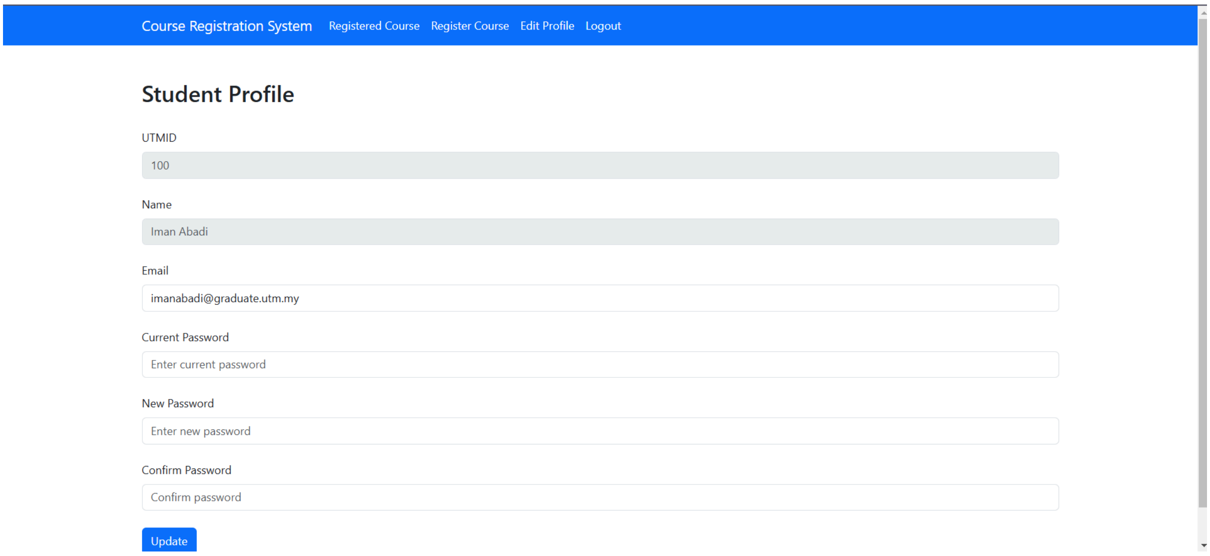
2023/2024-1

Submit

Search course... Search

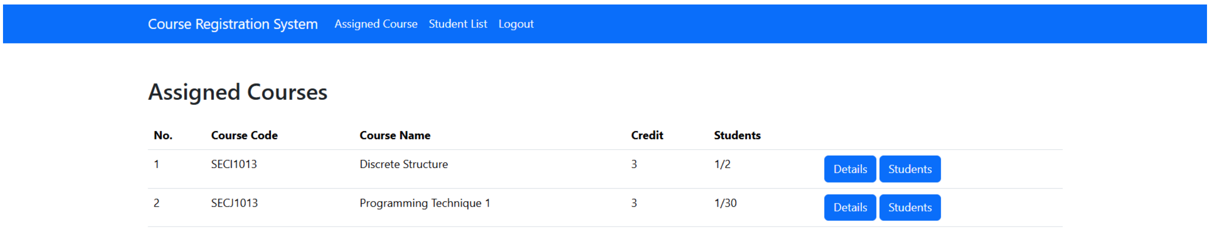
No.	Course Code	Course Name	Credit	Capacity	Semester	Lecturer	
1	SECI1013	Discrete Structure	3	1/2	2023/2024-1	Dr Shakir	Register
2	SECJ1013	Programming Technique 1	3	1/30	2023/2024-1	Dr Shakir	Register
3	SECP1513	Technology and Information System	3	0/30	2023/2024-1	Dr Alimi	Register
4	SECR1013	Digital Logic	3	1/30	2023/2024-1	Dr Alimi	Register

Edit Profile Page

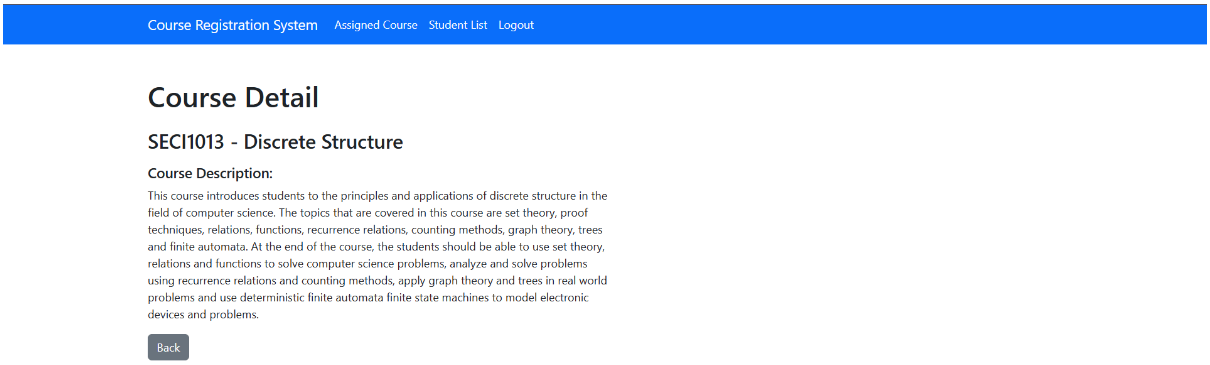


Lecturer

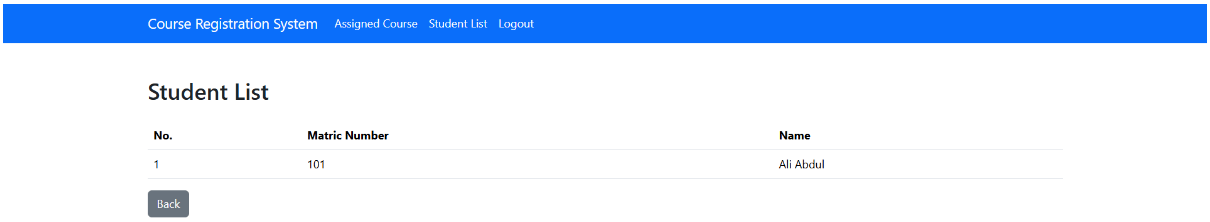
Assigned Course Page



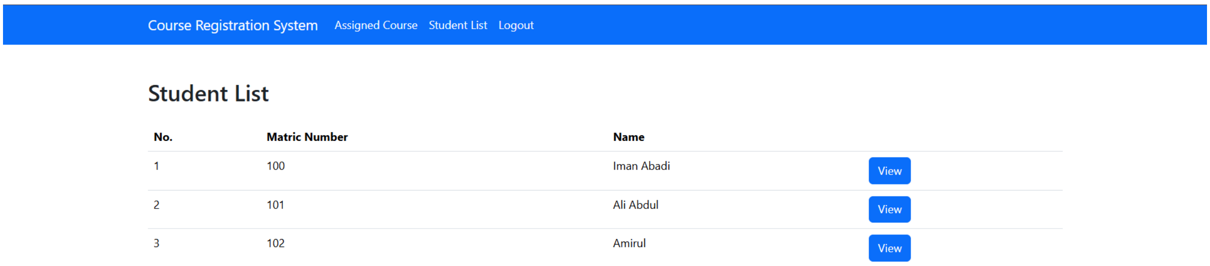
Course Detail Page



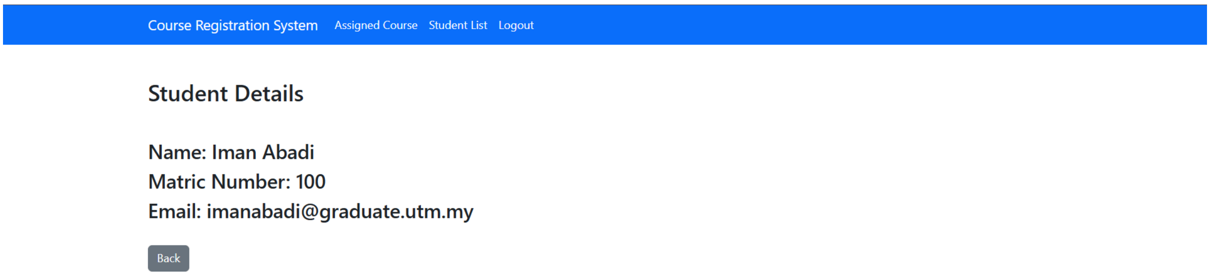
Student List of Assigned Course Page



Student List Page

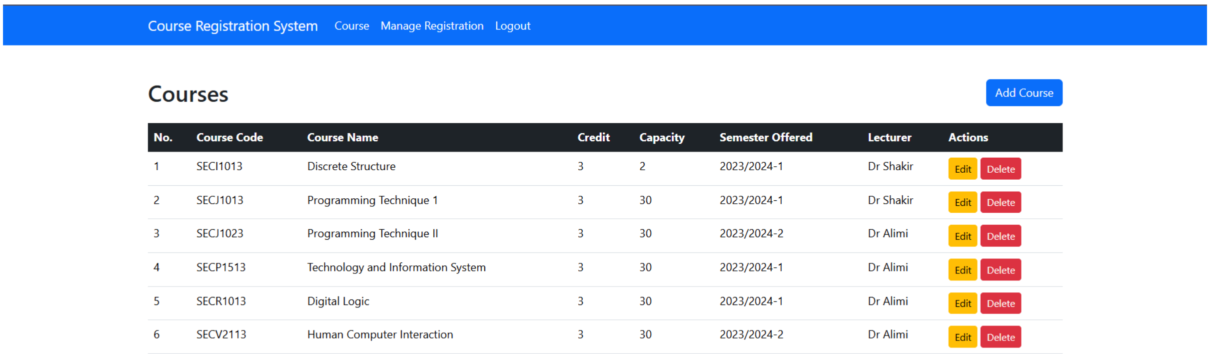


Student Detail Page



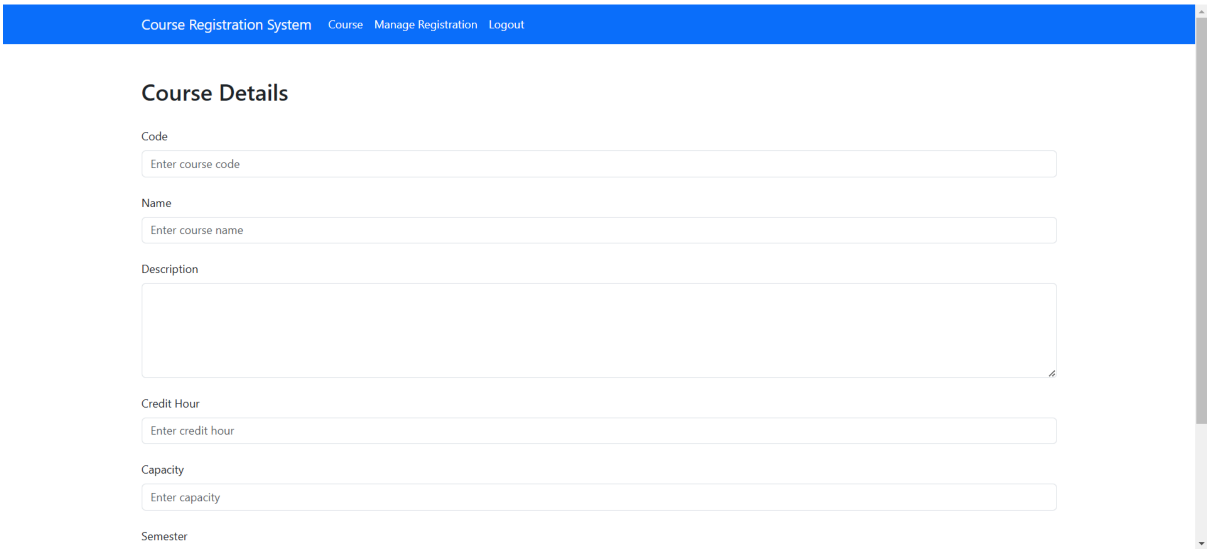
Admin

Admin Manage Course Page

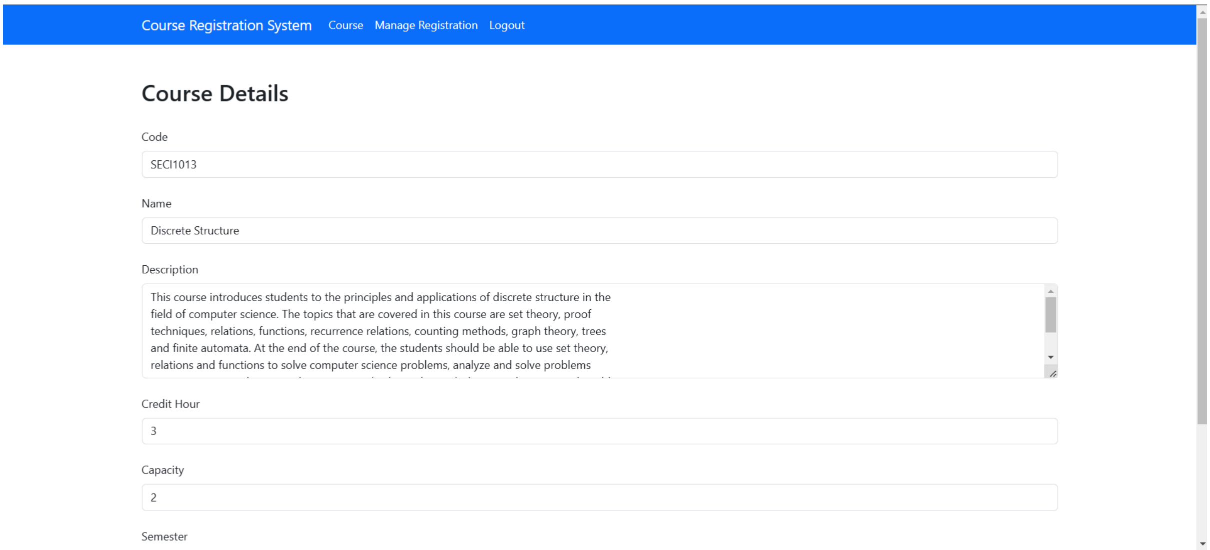


No.	Course Code	Course Name	Credit	Capacity	Semester Offered	Lecturer	Actions
1	SECI1013	Discrete Structure	3	2	2023/2024-1	Dr Shakir	Edit Delete
2	SECJ1013	Programming Technique 1	3	30	2023/2024-1	Dr Shakir	Edit Delete
3	SECJ1023	Programming Technique II	3	30	2023/2024-2	Dr Alimi	Edit Delete
4	SECP1513	Technology and Information System	3	30	2023/2024-1	Dr Alimi	Edit Delete
5	SECR1013	Digital Logic	3	30	2023/2024-1	Dr Alimi	Edit Delete
6	SECV2113	Human Computer Interaction	3	30	2023/2024-2	Dr Alimi	Edit Delete

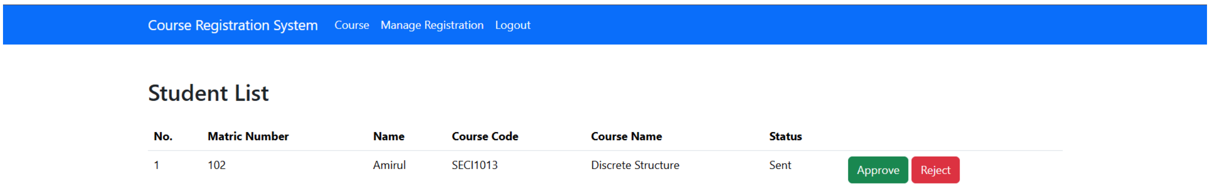
Admin Add Course Page



Admin Edit Course Page



Admin Amend Registration Page

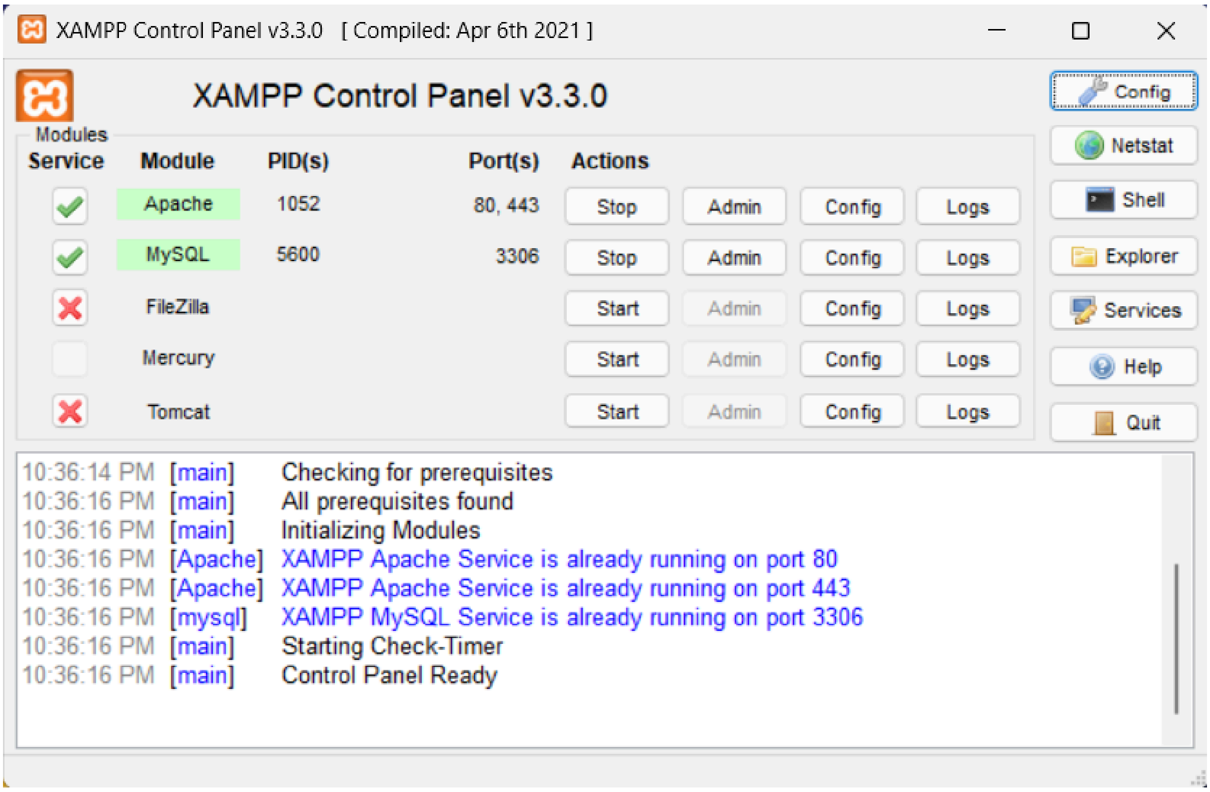


Localhost Setup

1. Download and install XAMPP Control Panel from

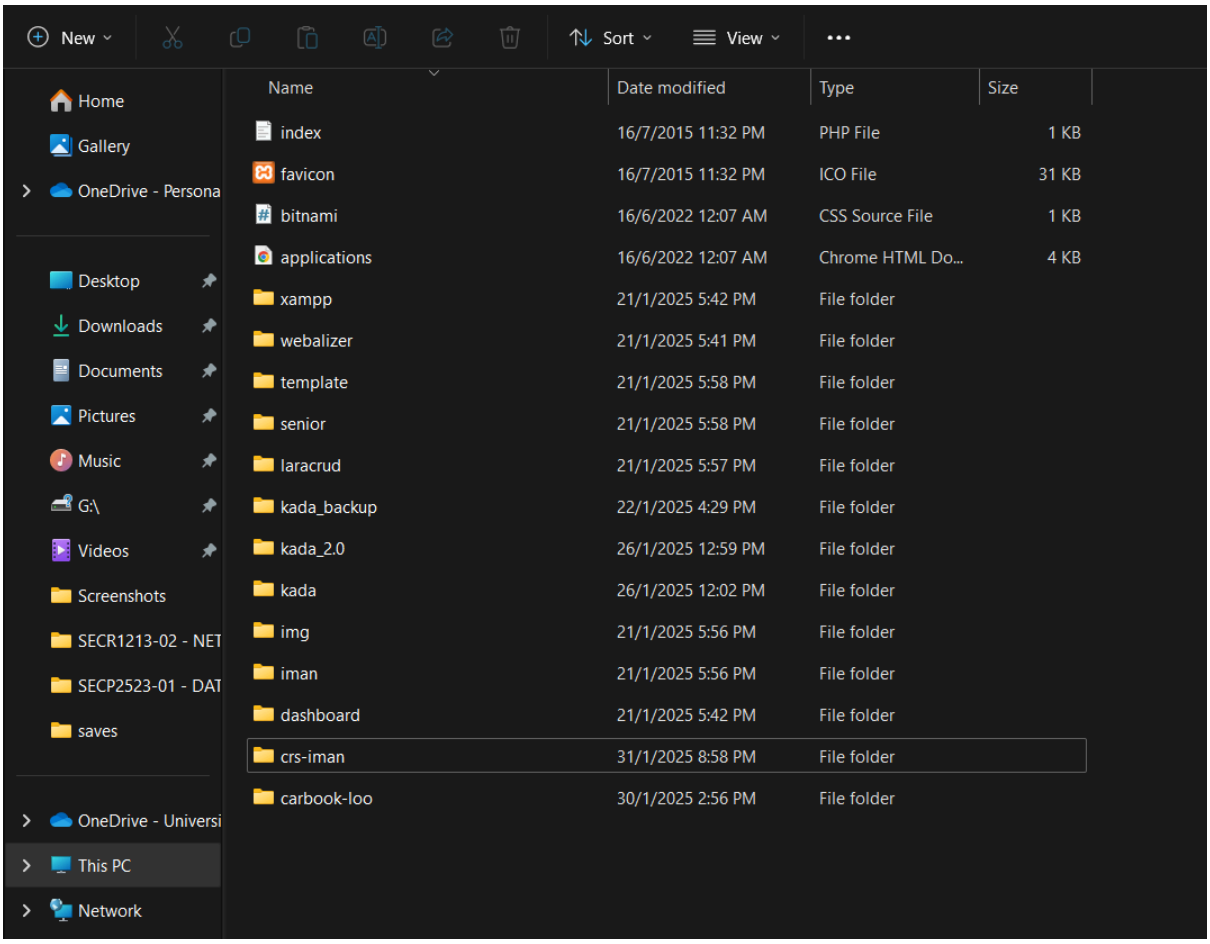
   https://www.apachefriends.org/index.html

2. Start Apache and MySQL.

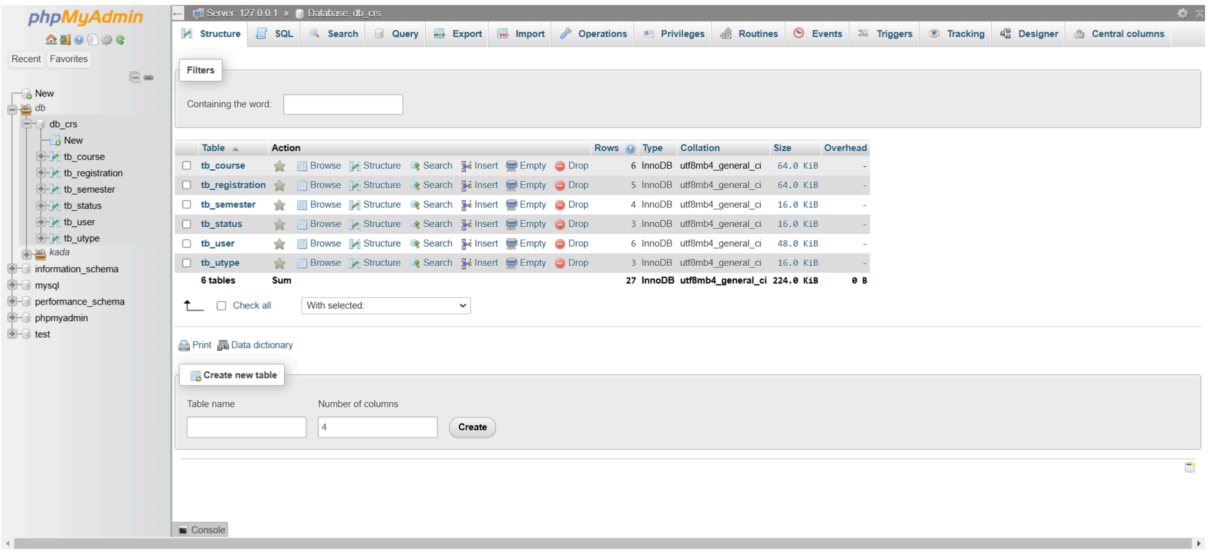


3. View phpMyAdmin by clicking 'Admin' button on 'Actions' section of MySQL module.

4. Extract the ‘crs-iman’ file into xampp > htdocs file path.



5. Create a new database with name ‘db_crs’



User Credentials

Student Accounts

UTMID: 100

Password: 1234

UTMID: 101

Password: 1234

UTMID: 102

Password: 1234

Lecturer Accounts

UTMID: 200

Password: 1234

UTMID: 201

Password: 1234

Admin Accounts

UTMID: 300

Password: 1234